Sharp unimodality for a fixed-parameter slice of the Stanley–Zanello family of Gaussian polynomial differences

[Author placeholder — to be completed upon owner approval]

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Abstract

For an integer $\ell \geq 5$ let

$$F_\ell(q) \;=\; \begin{bmatrix} \ell+8 \\ 8 \end{bmatrix}_q \;-\; q^{4\ell-18} \begin{bmatrix} 12 \\ 6 \end{bmatrix}_q .$$

We prove that F_ℓ is unimodal for every $\ell \geq 6$, and that the bound is best possible: F_5 is coefficientwise nonnegative but not unimodal. The family is a fixed-parameter slice $(k, b) = (8, 12)$ of the difference family of Stanley and Zanello, whose conjecture predicts this behaviour and is proved only for $k \leq 5$. The argument has three parts: an elementary Ferrers-diagram injection gives coefficientwise nonnegativity; symmetry about the common centre 4ℓ reduces unimodality to nineteen fixed-offset first-difference inequalities; and those inequalities are settled for all ℓ by combining an exact quasipolynomial description of restricted partition functions with a Newton forward-difference positivity certificate, plus one explicit finite verification for $6 \leq \ell \leq 839$. We state precisely which steps are theorems, which is a finite exact certificate, and what remains open.

1 Introduction and relation to Stanley–Zanello

Let $\begin{bmatrix} n \\ k \end{bmatrix}_q$ denote the Gaussian binomial coefficient, generating function for partitions inside a $k \times (n-k)$ rectangle. Stanley and Zanello [5] attached to parameters (k, m, b) a difference of two Gaussian polynomials

$$f(k,m,b)(q) \;=\; \begin{bmatrix} m \\ k \end{bmatrix}_q - q^{s(m)} \begin{bmatrix} b \\ k-2 \end{bmatrix}_q, \qquad s(m) = \tfrac{k(m-b)}{2} + b - 2k + 2,$$

admissible when $b \leq (km-4k+4)/(k-2)$ and $kb \equiv km \pmod 2$, and proved nonnegativity and unimodality completely for $k \leq 5$; their Conjecture 1.1(iv) predicts the same for every $k \geq 5$, all $m \gg_k 0$, and all admissible b except $b = (km-4k+2)/(k-2)$ when that is an integer, and they remark that for larger k "a new idea will most likely be necessary". Koutschan–Uncu–Wong [1] settled the conjecture for $k = 6$ and $k = 7$ with explicit exception tables; $k \geq 8$ is open.

In this paper we settle one explicit one-parameter subfamily: with $k = 8$, top parameter $m = \ell+8$, second-summand parameter $b = 12$, the shift is $s(m) = 4m-50 = 4\ell-18$ and the difference is exactly F_ℓ above. Admissibility holds precisely for $\ell \geq 5$ (the bound reads $12 \leq (8\ell+36)/6$),

the parity condition is automatic, and the exceptional-value condition $b = (km - 4k + 2)/(k - 2)$ becomes $m = 51/4$, never an integer, so this slice never lies on the exceptional locus. We do not claim the full $k = 8$ case, let alone the conjecture.

Unimodality of a *single* Gaussian polynomial is classical [2, 7], and strict unimodality is known [3, 6]; those results give first differences ≥ 1, which as we show in Section 6.2 is not enough here. Differences of Gaussian polynomials were studied earlier by Reiner and Stanton [4], whose conjecture the Stanley–Zanello family generalizes. Computer-algebra resolutions of related Stanley–Zanello families for small parameters appear in [1]; their parameter coverage does not include the slice treated here (see Section 9).

2 Definitions and notation

All polynomials lie in $\mathbb{Z}[q]$. For $P(q) = \sum_p c_p q^p$ we write $[q^m]P = c_m$ and set $c_m = 0$ outside $\{0, \ldots, \deg P\}$; this convention is used in every difference formula below. P is *unimodal* if $c_0 \leq \cdots \leq c_j \geq c_{j+1} \geq \cdots \geq c_D$ for some j, plateaus permitted; we never claim strict unimodality. $P \geq 0$ *coefficientwise* means $c_m \geq 0$ for all m. For $\ell \geq 5$ put

$$G_\ell(q) = \begin{bmatrix} \ell + 8 \\ 8 \end{bmatrix}_q, \quad a_\ell(n) = [q^n]G_\ell, \quad B(q) = \begin{bmatrix} 12 \\ 6 \end{bmatrix}_q, \quad b_j = [q^j]B, \quad F_\ell = G_\ell - q^{4\ell-18}B.$$

$\deg G_\ell = 8\ell$; $\deg(q^{4\ell-18}B) = 4\ell + 18 \leq 8\ell$ exactly when $\ell \geq 5$, which is the origin of the admissible range. Both summands are palindromic about degree 4ℓ, hence so is F_ℓ.

3 Main result

Theorem 1. *For every integer $\ell \geq 6$, $F_\ell(q)$ is unimodal.*

Theorem 2. *For every integer $\ell \geq 5$, $F_\ell(q) \geq 0$ coefficientwise.*

Theorem 3 (Sharpness)**.** *F_5 is coefficientwise nonnegative and not unimodal; explicitly $[q^{18..22}]F_5 = (15, 16, 15, 16, 15)$.*

Theorem 2 is proved by an elementary injection (Section 4); Theorem 1 by Theorems 5 and 10 (Sections 5–6); Theorem 3 by direct exact computation, which is legitimate because it is a single instance.

4 The Ferrers-diagram injection and nonnegativity

Identify $a_\ell(n)$ with the number of partitions $\lambda = (\lambda_1 \geq \cdots \geq \lambda_8 \geq 0)$ of n with $\lambda_1 \leq \ell$, and b_j with partitions of j inside a 6×6 rectangle. Padding a partition of $4\ell - 18 + j$ to exactly eight parts and adding $(1, 1, 1, 1, 0, 0, 0, 0)$ yields a partition of $4\ell - 14 + j$ inside the $8 \times (\ell + 1)$ box: the inequality $\lambda_4 \geq \lambda_5$ is preserved strictly, the degree rises by $4 = s(m + 1) - s(m)$, and the largest part rises by at most 1, exactly the box growth. Translation by a fixed vector is injective. Summing over the 37 offsets $0 \leq j \leq 36$ of the subtracted window gives the monotonicity statement below. The exact base check is at $\ell = 5$ ($m = 13$): the minimum window margin of F_5 is 1, attained at degree 2, so every window margin is at least $1 \geq 0$; one step further, at $\ell = 6$, the minimum window margin is 10, attained at degree 6. Propagating from the base

$\ell = 5$ gives Theorem 2. (An earlier draft of this package misstated the base margin at $\ell = 5$ as 10; the value 10 is the $\ell = 6$ margin. Both values are recomputed from the definitions in the accompanying code, and the verification program asserts them.)

Theorem 4 (Monotone window injection, even k [5])**.** *Let $k \geq 4$ be even and (m, b) admissible, and let $f(k, m, b) = \begin{bmatrix} m \\ k \end{bmatrix}_q - q^{s(m)} \begin{bmatrix} b \\ k-2 \end{bmatrix}_q$ with $s(m) = k(m-b)/2 + b - 2k + 2$. Then for every fixed offset $j \geq 0$,*

$$[q^{s(m)+j}] \begin{bmatrix} m \\ k \end{bmatrix}_q \;\leq\; [q^{s(m+1)+j}] \begin{bmatrix} m+1 \\ k \end{bmatrix}_q .$$

Consequently coefficientwise nonnegativity of $f(k, m, b)$ is monotone in m: once it holds at an admissible m_0 it holds for all admissible $m \geq m_0$, so eventual nonnegativity for fixed (k, b) reduces to a finite exact check.

Proof of Theorem 4. Let $k \geq 4$ be even and let (m, b) be admissible: $3 \leq k \leq m$, $3 \leq b \leq (km - 4k + 4)/(k - 2)$, and $kb \equiv km \pmod 2$; then $s(m) = k(m-b)/2 + b - 2k + 2$ is an integer, and admissibility of $(m+1, b)$ is automatic because k is even and the upper bound on b is nondecreasing in m. Represent $\begin{bmatrix} m \\ k \end{bmatrix}_q$ as generating partitions $\lambda = (\lambda_1 \geq \cdots \geq \lambda_k \geq 0)$ of the box $k \times (m-k)$, that is, with $\lambda_1 \leq m - k$, and degree $|\lambda|$. Define

$$\Phi(\lambda) = \big(\lambda_1 + 1, \ldots, \lambda_{k/2} + 1,\ \lambda_{k/2+1}, \ldots, \lambda_k\big).$$

Then $\Phi(\lambda)$ is a partition: the only adjacent pair affected is $(k/2,\, k/2+1)$, where $\lambda_{k/2} + 1 \geq \lambda_{k/2+1} + 1 > \lambda_{k/2+1}$; equivalently, the Ferrers diagram grows by exactly one cell in each of its first $k/2$ rows, which preserves the partition shape because row lengths are weakly decreasing. It lies in the $k \times (m+1-k)$ box because $\lambda_1 + 1 \leq (m-k) + 1 = (m+1) - k$. Its degree is $|\lambda| + k/2$, and $s(m+1) - s(m) = k/2$ since k is even. Hence Φ maps partitions of $s(m) + j$ in the $k \times (m-k)$ box into partitions of $s(m+1) + j$ in the $k \times (m+1-k)$ box; it is injective because it is translation by a fixed vector. Comparing coefficients yields $[q^{s(m)+j}] \begin{bmatrix} m \\ k \end{bmatrix}_q \leq [q^{s(m+1)+j}] \begin{bmatrix} m+1 \\ k \end{bmatrix}_q$ for every fixed offset $j \geq 0$.

For nonnegativity, outside the window $\{s(m), \ldots, s(m) + \deg \begin{bmatrix} b \\ k-2 \end{bmatrix}_q\}$ the coefficients of $f(k, m, b)$ equal those of $\begin{bmatrix} m \\ k \end{bmatrix}_q$ and are nonnegative; inside, the coefficient at offset j is $[q^{s(m)+j}] \begin{bmatrix} m \\ k \end{bmatrix}_q - b_j$, where $b_j = [q^j] \begin{bmatrix} b \\ k-2 \end{bmatrix}_q$ does not depend on m. The inequality above shows each window margin is nondecreasing along admissible m, so nonnegativity at one admissible m_0 — a finite check of $\deg \begin{bmatrix} b \\ k-2 \end{bmatrix}_q + 1$ integers — implies nonnegativity for all admissible $m \geq m_0$. For the slice $(k, b) = (8, 12)$ the base is $m_0 = 13$ ($\ell = 5$) with minimum window margin 1, as computed above. □

The argument requires k even; the odd-k analogue is not claimed. The injection controls nonnegativity only, and its technique is adjacent to the column-growth step of O'Hara's proof [2]; see Section 9 for the priority discussion.

5 Adjacent-difference reduction

Because F_ℓ is symmetric about 4ℓ, unimodality is equivalent to $[q^d]F_\ell \le [q^{d+1}]F_\ell$ for $0 \le d \le 4\ell - 1$. Writing $j = d - (4\ell - 18)$,

$$[q^{d+1}]F_\ell - [q^d]F_\ell \;=\; \big(a_\ell(d+1) - a_\ell(d)\big) - \big(b_{j+1} - b_j\big).$$

For $j \le -2$ both b-indices are negative and the right-hand side equals $a_\ell(d+1) - a_\ell(d) \ge 0$ by unimodality of G_ℓ, since $d + 1 < 4\ell$. For $j \ge 18$ symmetry applies. Exactly $j = -1, 0, \ldots, 17$ carry conditions. The endpoint $j = -1$ is not free: $b_0 - b_{-1} = 1$.

Theorem 5 (Reduction)**.** *For every $\ell \ge 6$, F_ℓ is unimodal if and only if*

$$a_\ell(4\ell - 18 + j + 1) - a_\ell(4\ell - 18 + j) \;\ge\; Db_j \qquad (j = -1, 0, \ldots, 17),$$

where $Db_j = b_{j+1} - b_j$ is the vector displayed in Table 1.

j	-1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Db_j	1	0	1	1	2	2	4	2	5	4	6	4	7	3	6	3	4	0	3

Table 1: The nineteen required lower bounds, computed from $\begin{bmatrix}12\\6\end{bmatrix}_q$ in the accompanying code rather than tabulated by hand. The interior maximum is $Db_{11} = 7$; the binding case for large ℓ is $j = 17$, the step immediately before the centre.

6 Quasipolynomial certification

Lemma 6. *Let $p(n) = [q^n]\prod_{i=1}^{8}(1-q^i)^{-1}$ with $p(n) = 0$ for $n < 0$. Then p is a quasipolynomial in n of degree at most 7 and period dividing $840 = \mathrm{lcm}(1, \ldots, 8)$, valid for every $n \ge 0$.*

Proof. The generating function $P(q) = 1/D(q)$ with $D(q) = \prod_{i=1}^{8}(1 - q^i)$ is a *proper* rational function: the numerator is 1 and $\deg D = 1 + \cdots + 8 = 36 > 0$. Its poles are the roots of unity ζ of order dividing $840 = \mathrm{lcm}(1, \ldots, 8)$, and the multiplicity of ζ as a pole is $m(\zeta) = |\{1 \le i \le 8 : \zeta^i = 1\}| \le 8$, attained only at $\zeta = 1$. Because the fraction is proper, its partial fraction decomposition over $\mathbb{C}$ has *no polynomial part*: it is the exact identity of rational functions

$$\frac{1}{D(q)} \;=\; \sum_{\zeta^{840}=1} \sum_{r=1}^{m(\zeta)} \frac{c_{\zeta,r}}{(1 - q/\zeta)^r}, \qquad c_{\zeta,r} \in \mathbb{C},$$

summed over all 840-th roots of unity. Expanding each term as a geometric-type series, $[q^n](1 - q/\zeta)^{-r} = \binom{n+r-1}{r-1}\zeta^{-n}$, an identity of formal power series valid for *every* $n \ge 0$. Hence for all $n \ge 0$,

$$p(n) = \sum_{\zeta^{840}=1} \sum_{r=1}^{m(\zeta)} c_{\zeta,r} \binom{n + r - 1}{r - 1} \zeta^{-n},$$

and since $\zeta^{-840} = 1$, on each residue class $n \equiv \rho \pmod{840}$ the factors ζ^{-n} are constant, so $p(\rho + 840t)$ is a polynomial in t (equivalently in n) of degree at most $\max_\zeta m(\zeta) - 1 = 7$. Grouping

conjugate summands shows the class polynomials have rational coefficients; they take integer values because p does. No "sufficiently large n" qualification appears anywhere: the only finitely-supported contributions that ordinary asymptotic quasipolynomial statements need are absent here precisely because the numerator is constant and the decomposition carries no polynomial part. □

Lemma 7 (Inclusion–exclusion)**.** $a_\ell(n) = \sum_{S\subseteq\{1,\dots,8\}} (-1)^{|S|}\, p\big(n - |S|\ell - \sigma(S)\big)$, $\sigma(S) = \sum_{i\in S} i$.

Proof. Expand $\prod_{i=1}^{8}(1-q^{\ell+i})$ in $\begin{bmatrix}\ell+8\\ 8\end{bmatrix}_q = \prod_{i=1}^{8}(1-q^{\ell+i})\prod_{i=1}^{8}(1-q^i)^{-1}$ and extract coefficients. □

For the offsets at hand $n = 4\ell - 18 + j$ with $j \le 17$, so the argument of p belonging to a subset S with $|S| = 4$ is at most $-1-10 = -11 < 0$, and for $|S| \ge 5$ it is at most $-\ell - 11$. Hence:

Corollary 8. *For $\ell \ge 840$ and each offset j, the margin $Q_{j,r}(t) := a_\ell(4\ell - 18 + j + 1) - a_\ell(4\ell - 18 + j) - Db_j$ with $\ell = r + 840(t+1)$, $0 \le r < 840$, is a polynomial in t of degree at most 7, valid for all integer $t \ge 0$.*

Lemma 9 (Newton positivity)**.** *If a real polynomial Q of degree at most 7 satisfies $\Delta^i Q(0) \ge 0$ for $i = 0, \dots, 7$, where Δ is the forward difference, then $Q(t) \ge 0$ for every integer $t \ge 0$.*

Proof. Newton's expansion $Q(t) = \sum_{i=0}^{7} \binom{t}{i} \Delta^i Q(0)$ with $\binom{t}{i} \ge 0$ for integer $t \ge 0$. □

Theorem 10 (Certificate)**.** *For every $j \in \{-1, \dots, 17\}$ and every $r \in \{0, \dots, 839\}$, the eight Newton coefficients $\Delta^i Q_{j,r}(0)$, $i = 0, \dots, 7$, are nonnegative. Consequently the nineteen inequalities of Theorem 5 hold for all $\ell \ge 840$.*

The theorem's hypothesis is a finite assertion about $19 \cdot 840 = 15{,}960$ integer octuples, produced and validated by the accompanying code (Section 7). Two features are worth recording. First, the sixth and seventh Newton coefficients vanish identically in every class, so the true degree is at most 5; the degree-7 bound is used only to size the certificate. Second, the smallest tail margin is 3,552,106,826, so the tail is far from sharp: all the tension in Theorem 1 lives in the finite base $6 \le \ell \le 839$, where the minimum margin is 1, attained at $(\ell, j) = (6, 16)$, and where the central surplus equals exactly 4 against the required 3 at $\ell = 6, 7$.

6.1 Why this is a certificate and not experimental evidence

Corollary 8 and Lemma 9 together turn the computation into a proof of a statement about *all* $\ell \ge 840$: the computation only verifies the hypotheses of a theorem about polynomials, over an explicitly enumerated finite set. No interpolation, no floating point, and no asymptotic fitting is involved; the values are integers obtained from the definitions, and the verification package recomputes the complete Newton vectors of every obligation through a second independent code path (dense truncated Gaussian multiplication), so a value-tampered certificate cannot pass.

6.2 The obstruction to an injection-only proof

The injection of Section 4 can be sharpened to an exact identity. Adding 1 to the largest part maps partitions of n with $\lambda_1 \le \ell-1$ bijectively onto partitions of $n+1$ with $\mu_1 \ge \mu_2+1$, whence

$$a_\ell(n+1) - a_\ell(n) \;=\; \#\{\mu \vdash n+1 \text{ in the } 8\times\ell \text{ box} : \mu_1 = \mu_2\} \;-\; \#\{\lambda \vdash n \text{ in the box} : \lambda_1 = \ell\}.$$

Both terms are, at fixed offset, degree-6 quasi-polynomials in ℓ, and their degree-6 leading terms cancel exactly while the degree-5 term survives: the difference is of degree exactly 5 on each residue class modulo 840—precisely what the certificate records (the fifth Newton coefficient is nonzero in all 15,960 classes and the sixth and seventh vanish identically). The mechanism is the palindromic symmetry $a_\ell(m) = a_\ell(8\ell - m)$: the window sits at the centre of the coefficient array, so the continuous slice-volume expansion in powers of $1/\ell$ is even about the centre and its first difference contributes no degree-7 or degree-6 term, with leading smooth term proportional to $(c+1)^2 - c^2 = 2c+1$ at degree ℓ^5 (the lower-degree periodic contributions are unrestricted but cannot raise the degree above 5); exact rational interpolation on sampled residue classes confirms that the degree-7 and degree-6 terms of $a_\ell(4\ell + c)$ agree with those of $a_\ell(4\ell + c + 1)$ (hence cancel in the difference) and that the surviving degree-5 coefficient is proportional to $2c + 1$, so it is nonzero for every integer offset c. A residual of degree 5 must be compared against absolute constants (3 and 7). An injection yields ≥ 0; here one needs ≥ 3, and equality is nearly attained (true value 4 at $\ell = 6$). Box symmetries (conjugation, complement) supply no surplus set of the required size. This is the precise sense in which the elementary mechanism does not extend to Theorem 1, and why a quantitative input of a different kind is needed.

7 Finite certificate and reproducibility

The companion directory `verification/` contains three standard-library Python programs and a recorded expectation file: `generate_certificate.py` builds the 15,960-obligation certificate from the definitions (offset constants derived from $\begin{bmatrix}12\\6\end{bmatrix}_q$ inside the program, never read from a table); `verify_certificate.py` re-derives those constants, checks structural completeness (no missing or duplicated obligations), checks nonnegativity of all Newton coefficients, checks the degree drop, and recomputes the full eight-coefficient Newton vector of *every* one of the 15,960 obligations through an independent dense code path (plus a seeded sample through a third route), failing non-zero on a missing, truncated, or value-tampered certificate; `independent_check.py` verifies the theorem statements themselves by constructing F_ℓ completely for $\ell = 5, 6, 7$, sweeping $6 \le \ell \le 80$, testing large parameters, and confirming the equivalence of Theorem 5 against direct unimodality for $5 \le \ell \le 60$. Reference timings on a current Fedora/x86_64 machine with Python 3.15: generation about one second, full verification about sixty-five seconds (dominated by the 15,960-class dense recomputation), independent checks under two seconds—so an outside reviewer can reproduce everything without specialized hardware. A SHA-256 manifest covers the package; it certifies byte integrity, not mathematical correctness.

8 The failure at the preceding parameter

At $\ell = 5$ the window touches low degrees and the cancellation is no longer favourable: exact computation gives central coefficients $(15, 16, 15, 16, 15)$ at degrees $18, \ldots, 22$, so the sequence

rises, falls, rises, falls, and unimodality fails while nonnegativity survives. This single instance is the content of Theorem 3; it also explains the shape of the finite base: the margin is smallest precisely at the bottom of the admissible range.

9 Relationship to previous work

Unimodality of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is due to O'Hara [2], with Zeilberger's shorter proof [7]; the KOH decomposition underlying both is the standard structural tool. Strict unimodality results of Pak–Panova [3] and Zanello [6] give lower bounds of magnitude 1 on central first differences, insufficient for Theorem 1, where 3 is required at $j = 17$ and 7 at $j = 11$. Reiner–Stanton [4] conjectured unimodality of symmetric differences of Gaussian polynomials, of which the Stanley–Zanello family is a generalization; their proved cases do not cover our slice. Koutschan, Uncu and Wong [1] resolved, using cylindrical algebraic decomposition and split points, two additional cases of the Stanley–Zanello conjecture — lower parameters 6 and 7, with explicit exception tables — and their d-strict monotonicity framework for single Gaussian polynomials stops computationally at $m = 7$; the $k = 8$ slice treated here lies outside their proved coverage, as recorded with explicit hypothesis/conclusion comparisons in the accompanying verification package. Quasipolynomiality of restricted partition functions is classical (Sylvester; denert–Desarmens and Sturmfels in the vector-partition setting), and positivity certificates based on finite exact computations are standard in computer-assisted proofs; we claim no novelty for those tools as such.

10 Limitations and open questions

1. Scope. Theorem 1 is a single fixed-parameter slice. It is not the full $k = 8$ case of [5], and it says nothing about odd k.

2. Certificate versus conceptual proof. Positivity of the margin polynomials is established by a 15,960-class exact certificate. A hand-checkable argument — for instance a closed form for the residual quintic together with a sign analysis, or a combinatorial surplus injection of size 3 at $j = 17$ — is missing and is the most natural next target.

3. Degree drop. The observed drop from degree 7 to degree 5 is uniform across all classes but is presently recorded only as a computation; a structural explanation (cancellation between the flat-top and right-edge layers of Section 6.2) is open.

4. Sharpness of the tail. The tail margins are enormous while the finite base has margin 1; no optimality is claimed for 840 as the junction between certificate and finite base. In fact the truncation of Lemma 3 holds for every $\ell \geq 1$ (the offending arguments are ≤ -11 uniformly), while polynomiality of the surviving terms needs only $\ell \geq 25$; a sharper packaging would certify $\ell \geq 25$ per residue class and shrink the finite verification to $6 \leq \ell \leq 24$. The uniform parametrization used here trades that tightening for a single polynomial per residue class.

5. Verification trust. The certificate is machine-verified exact-integer computation in a standard-library implementation; it has been reproduced by two independent implementations, but no interactive proof assistant formalization exists.

6. No external peer review has been performed.

Reproducibility statement

All computations are exact integer arithmetic in Python 3 with the standard library only, deterministic and randomized-sample-checked through a second code path; commands, reference outputs, software versions, and file hashes are recorded in `verification/EXPECTED_RESULTS.md`; the complete runnable package is distributed as `math06-verification.zip`.

References

[1] Christoph Koutschan, Ali K. Uncu, and Elaine Wong. A unified approach to unimodality of Gaussian polynomials. pages 434–442, 2023. doi: 10.1145/3597066.3597113. Section 4 resolves the Stanley–Zanello conjecture for lower parameters 6 and 7 (with explicit exception tables) via cylindrical algebraic decomposition; the $k = 8$ slice treated in the present paper is outside that coverage.

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